# Facebook

System Design - Proposal

# Team Members

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# Feature Categories

## Core Features

- Posts System
- Comments System
- User Profiles

## Intelligent Features

- News Feed Recommendation
- Friend/Group Suggestions

## Admin Features

- Group/Page Management
- Content Moderation
- Analytics Dashboard

# 1. Posts System

## CRUD Operations

- CREATE: Text, photos, videos on timeline
- READ: View posts on News Feed, profiles, pages
- UPDATE: Edit content, change privacy settings
- DELETE: Remove posts or move to recycle bin

## Enhanced Features

- Privacy Controls: Public / Friends / Custom visibility
- Post Reactions
- Engagement Tracking: Track views, reactions, shares for ranking
- Visibility Algorithm: Basic post ranking for News Feed

# 2. Comments System

## 1

### CRUD Operations

- CREATE: Post replies to posts, photos, or other comments
- READ: View comment threads and nested replies
- UPDATE: Edit text of existing comment after posting
- DELETE: Delete own comments or hide on own posts

## 2

### Enhanced Features

- Nested Comments: Multi-level reply chains with proper threading
- Comment Sorting: Sort by Top, Most Recent, or Most Relevant
- Mention Functionality: Tag users using @username with notifications
- Reaction Support: React to comments with emoji reactions

# 3. Post Ranking Algorithm

- Recency score (time-based decay)
- Engagement score (likes, comments, shares)
- User interest matching
- Connection strength weighting

Filtering & Personalization:

1. Filter by friend connections
2. Consider user preferences
3. Privacy setting validation
4. Content diversity balancing

Implementation Approach:
Simple weighted scoring system combining recency, engagement metrics, and user relationships.

# 4. Friend/Group Suggestion Engine

## Intelligent Recommendation Component

Friend Suggestions:

- Mutual friends algorithm
- Common interests detection
- Shared group memberships
- Network proximity scoring
- "People You May Know" feature

Group Suggestions:

- Interest-based matching
- Friend membership overlap
- Activity level consideration
- Category-based filtering
- Geographic relevance

Implementation Approach:
Graph-based algorithms analyzing social connections and shared attributes. It requires graph traversal, similarity scoring, and relationship-based recommendations without requiring machine learning.

# 5. Administrative Control Panel (Group/Page)

## Component for Access Control Based on Roles

Moderation of Submitted Content:

1. Approve or deny submitted posts
2. Examine content that has been reported
3. Conceal or remove content violations
4. Establish filters for content management

Management of Community Members:

1. Include or exclude members from the group
2. Designate roles for moderators
3. Prohibit or silence users from activity
4. Queue for approving new members

Data Analysis Dashboard:

1. Metrics on the reach of posts
2. Statistics on user engagement
3. Monitoring of growth patterns
4. Count of currently active members

# 6. User Profiles (Supporting)

## 1 Core Features

- Basic Information: Name, bio, profile picture, cover photo
- Interest Tags: User-defined interests for recommendation system
- Activity Feed: Recent posts and interactions display
- Friends List: View and manage friend connections

## 2 Privacy Settings

- Profile Visibility: Control who can view profile information
- Friend List Privacy: Show/hide friend connections
- Activity Visibility: Control timeline post visibility
- Search Settings: Manage discoverability by others